

Kewei Portfolio

A Digital Portfolio Website

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Date: September 2026

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1. Declaration and Transparency

Generative AI Transparency Statement

AI Usage Level: 4 — AI Task Completion, Human Evaluation

Tool	Usage Level	Precise Usage
Joy AI (JoyCode)	Level 4	Generated project scaffolding (Vite + React + TypeScript setup), directory structure, boilerplate component code, and CSS styling. All AI-generated code was reviewed and understood by the student.

2. Abstract

A portfolio website built with React, TypeScript, and Vite. It has nine pages, uses CSS Modules and CSS variables for styling, and includes a custom chat hook. Technologies are kept simple to stay lightweight. This report covers the discovery, design, implementation, and evaluation of the project.

3. Discovery and Narrative Description

3.1 Overview

A React-based portfolio with nine pages covering background, skills, projects, blog, and messaging.

3.2 Feature List

#	Feature	Page/Route	Phase	Description
1	Home Page	/	1	Intro and feature cards
2	About Me	/about	1	Personal background and interests
3	Education	/education	1	Education and work timeline
4	Professional Knowledge	/knowledge	1	Skills grouped by category
5	Photo Gallery	/gallery	1	Photo grid of projects and events
6	Video Gallery	/videos	1	Embedded YouTube videos
7	Navigation Bar	—	1	Top navigation with active styling
8	Client-side Routing	—	1	React Router page navigation
9	CSS Styling	—	1	CSS Modules and theme variables
10	Blog	/blog, /blog/:id	2	Articles with Markdown detail
11	Instant Messaging	/messaging	2	Chat with auto-reply bot
12	README	/readme	2	Project overview page
13	Dark Mode	—	2	Auto dark/light theme switch
14	Responsive Design	—	2	Mobile-friendly layout
15	Error Handling	—	2	404 page and Error Boundary

3.3 Narrative Description

The portfolio is a single-page application built with React, TypeScript, and Vite. It has nine pages linked through a top navigation bar. Phase 1 covers six static content pages, routing, and CSS theming. Phase 2 adds interactive features — blog, messaging, and dark mode — plus responsive design and error handling.

4. Iterative Project Plan

4.1 Phase 1 — Core Structure and Static Pages (Sep 15 – Oct 6)

Iteration	Duration	Deliverables
1.1	Sep 15–28	Project setup, routing, styling, Navbar, Home page
1.2	Sep 29 – Oct 6	About, Education, Knowledge, Gallery, Videos pages

4.2 Project Peer Review (Oct 6 – Oct 24)

Iteration	Duration	Deliverables
—	Oct 6–24	Peer review of Phase 1

4.3 Phase 2 — Interactive Features and Polish (Oct 24 – Nov 14)

Iteration	Duration	Deliverables
2.1	Oct 24 – Nov 7	Blog, Messaging, README pages
2.2	Nov 8–14	Dark mode, responsive design, final polish

5. Tools and Techniques

5.1 Development Stack

Category	Tool / Technology	Version	Purpose
Framework	React	19.2.8	Build UI components
Language	TypeScript	6.0.2	Type checking
Build Tool	Vite	8.3.0	Dev server and bundling
Routing	React Router DOM	7.18.4	Page navigation
Markdown	react-markdown	10.1.0	Render blog content
Linting	ESLint	10.10.0	Code smell detection
Package Manager	npm	—	Install and run packages
Version Control	Git	—	Track code changes
Repository	GitHub	—	Host code remotely

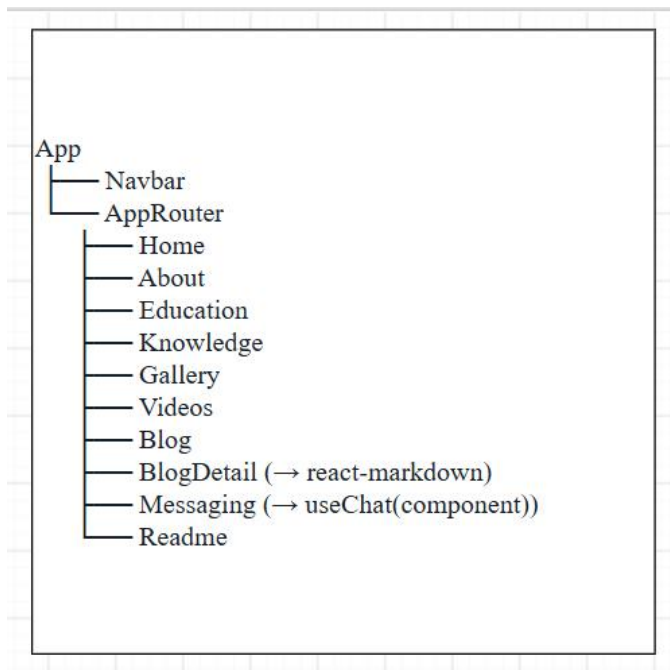
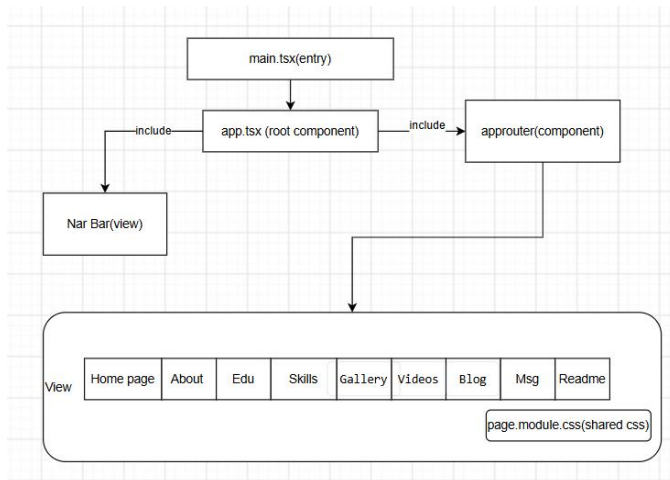
5.2 Techniques

Technique	Application
CSS Modules	Scoped styling per component
CSS Custom Properties	Shared color and spacing variables
Custom React Hooks	Separate logic from UI (useChat)
Dark Mode	Auto theme switch via prefers-color-scheme
Static Data Separation	Data constants separated from rendering

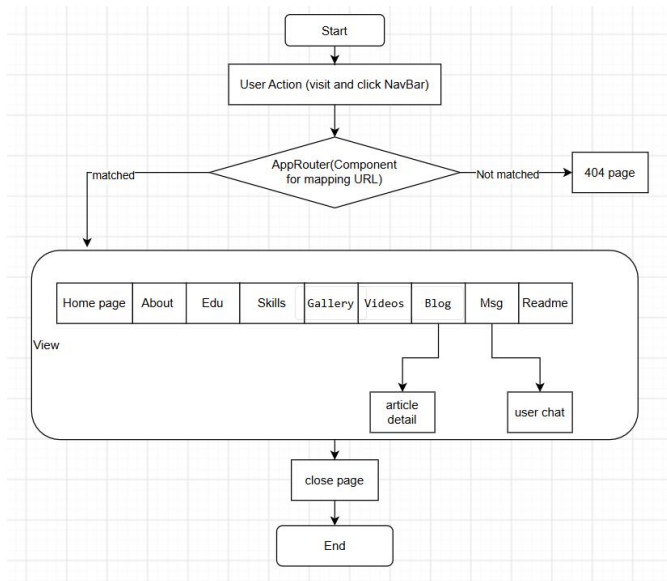
6. Design Discussion

6.1 Architecture and Component Diagram

main.tsx → App.tsx → BrowserRouter → Navbar + AppRouter → Pages → Hooks (useChat) → Styles (CSS Modules + Variables).



6.2 Control Flow Diagram



6.3 Design Decisions

1. **CSS Variables in index.css:** Shared theming and auto dark mode.
2. **useChat Hook:** Chat logic separated from UI.
3. **Static Data Separation:** Data constants kept apart from rendering.
4. **react-markdown for Blog:** Render rich text without a CMS.

7. Code Listing

7.1 GitHub Repository

<https://github.com/franklinprivateuse-jpg/assignment>

7.2 Project Structure

```
kewei-portfolio/
├── public/
│   ├── favicon.svg
│   └── icons.svg
├── src/
│   ├── app/
│   │   └── AppRouter.tsx          # Route configuration
│   ├── assets/
│   │   ├── background_picture.png
│   │   ├── hero.png
│   │   ├── react.svg
│   │   └── vite.svg
│   ├── components/
│   │   └── navbar/
│   │       ├── Navbar.tsx
│   │       └── Navbar.module.css
│   ├── hooks/
│   │   └── useChat.ts            # Chat logic hook
│   ├── pages/
│   │   ├── page.module.css      # Shared page layout
│   │   ├── about/
│   │   ├── blog/
│   │   │   ├── Blog.tsx
│   │   │   ├── BlogDetail.tsx
│   │   │   ├── types.ts
│   │   │   └── *.module.css
│   │   ├── education/
│   │   ├── gallery/
│   │   ├── home/
│   │   ├── knowledge/
│   │   ├── messaging/
│   │   ├── readme/
│   │   └── videos/
│   ├── App.tsx                  # Root component
│   ├── index.css                # Global styles & CSS variables
│   └── main.tsx                 # Entry point
├── index.html
├── package.json
├── tsconfig.app.json
├── tsconfig.json
├── tsconfig.node.json
├── vite.config.ts
└── eslint.config.js
```

8. Innovation

8.1 Architecture Innovation

Page-based layout with co-located CSS Modules — no global state, clear separation, Markdown blog without a CMS.

9. Critique

9.1 Strengths

- Clean separation of pages, components, hooks, and styles.
- CSS variables with auto dark mode.
- useChat hook separates logic from UI.

9.2 Weaknesses

- No tests at all.
- No 404 page or Error Boundary.
- No mobile menu (Hamburger Menu) for navbar.

9.3 Enhancement Suggestions

- Add tests for the useChat hook and key pages.
- Add a 404 page and Error Boundary.
- Add a hamburger menu for mobile.

10. Reflection

10.1 Learning Outcomes

- Using CSS variables and CSS Modules for styling.
- Separating logic from UI with custom hooks.
- Page routing with React Router.

10.2 Challenges Faced

- Balancing features and quality in limited time.
- Too many navbar links for mobile screens.

11. References

[1] GeeksforGeeks (2025). Difference Between TypeScript and JavaScript. Available at: <https://www.geeksforgeeks.org/typescript/difference-between-typescript-and-javascript/> (Accessed: September 2026).

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